

1 System Model and Practical Challenges in Over-the-Air (OTA)

A system that takes advantage of OTA can be modeled in different ways, also depending on the use case and environment. Most layouts involve a number of End Device (ED)s that transmit to a Fusion Center (FC) or Edge Server (ES). The channel model is also an important aspect, as it can be affected by fading, noise, and path loss. In addition to the system model, there are also practical challenges that can affect the reliability of OTA, such as synchronization errors and waveform design. To enhance the performance of OTA, optimization strategies can be employed, such as power management and transceiver design.

1.a System Architecture and Channel Model

In this section the mathematical model of a network consisting of three EDs will be presented. These devices are deployed at an agricultural field and take advantage of OTA in order to transmit to the FC, by using Pulse Amplitude Modulation (PAM). The physical environment has multipath characteristics and an impulse response with a span of three taps.

- *Modeling the Network*: Description of EDs transmitting to an FC, possibly involving multiple antennas (Multiple-Input Multiple-Output (MIMO)) [1], [2], [3], [4], [5].
 - **Sensor 1**: $x_1[n] = [2, -1, 1]$
 - **Sensor 2**: $x_2[n] = [1, 2, -1]$
- *Wireless Channel Model*: Consideration of channel impairments, including fading (e.g., Rayleigh fading), noise (Additive White Gaussian Noise (AWGN)), and path loss [4], [6], [7].

The channel impulse response is given as: $h[n] = [1.0, 0.5, -0.2]$

(Here zero noise is assumed)

To calculate the ideal target function of the OTA aggregated sequence, $s[n]$, the ideal flat channel without multipath is assumed as: $h_{ideal}[n] = [1]$.

$$\begin{aligned} s[n] &= x_1[n] + x_2[n] \Rightarrow \\ s[n] &= [3, 1, 0] \end{aligned} \tag{1}$$

1.b Waveform Design and Intersymbol Interference in Nomographic Computations

To transcend idealized discrete-time models, it is imperative to analyze the Over-the-Air (OTA) computation architecture in the continuous-time domain, incorporating the physical constraints of pulse shaping and channel memory.

According to Kolmogorov's superposition theorem, a multivariate target function $f : \mathbb{R}^K \rightarrow \mathbb{R}$ can be represented nomographically. Assuming the Fusion Center (FC) necessitates the geometric mean of the sensor observations $v_k[m]$, the requisite pre-processing and post-processing functions are designated as:

$$\phi_k(v_k) = \ln(v_k), \quad \psi(y) = \exp\left(\frac{y}{K}\right) \tag{2}$$

The End Devices (EDs) emit the pre-processed symbols $s_k[m] = \phi_k(v_k[m])$ via Pulse Amplitude Modulation (PAM). Employing a Root Raised Cosine (RRC) filter $g(t)$ to strictly bound the spectral occupancy, the transmitted waveform of the k -th device is formulated as:

$$u_k(t) = \sum_{m=-\infty}^{\infty} s_k[m]g(t - mT_s) \tag{3}$$

Propagating through a multiple-access channel characterized by a fractional multipath delay profile $h(t) = \delta(t) + \alpha_1 \delta(t - \tau)$, the aggregated signal arrives at the FC. Upon traversing the matched filter $g(-t)$ and undergoing sampling at $t = nT_s$, the discrete sequence emerges as:

$$y[n] = \sum_{k=1}^K s_k[n] + \alpha_1 \sum_{k=1}^K \sum_{m=-\infty}^{\infty} s_k[m] p(nT_s - mT_s - \tau) + w[n] \quad (4)$$

where $p(t)$ denotes the Raised Cosine Nyquist pulse, and $w[n]$ signifies the filtered noise sequence.

Equation (3) explicitly segregates the desired OTA superposition (the primary term) from the severe Inter-Symbol Interference (ISI) induced by the fractional delay τ disrupting the zero-crossing alignment of $p(t)$.

To evaluate the degradation imposed by this fractional delay, an extensive computational model was constructed in MATLAB. The simulation iterates the excess bandwidth parameter (roll-off factor α) to quantify the attenuation of the ISI tails and its subsequent impact on the Mean Squared Error (MSE).

1.c Impact of Practical Impairments on OAC Reliability

In order to calculate the received signal with ISI at the FC, the linear property of the channel can be taken advantage of.

$$\begin{aligned} y[n] &= s[n] * h[n] \Rightarrow \\ y[n] &= \sum_{m=0}^n s[m] h[n-m] \Rightarrow \\ y[n] &= [3.0, 2.5, -0.1, -0.2, 0]; \end{aligned} \quad (5)$$

It is obvious that because of the ISI the three symbols of 1 are spread to the five symbols of 5.

To calculate the Mean Squared Error (MSE) at each sample the comparison of the actual received signal to the ideal target function shall be made.

$$\begin{aligned} e[n] &= y[n] - n[n] \Rightarrow \\ e[n] &= [0.0, 1.5, -0.1, -0.2, 0.0] \\ MSE &= \frac{e[n]}{5} \Rightarrow \\ MSE &= 0.46 \end{aligned} \quad (6)$$

1.c.i Synchronization Errors

Time, Frequency, and Phase Offsets (TO, CFO, PO): The detrimental impact of imperfect time-of-arrivals and phase alignment on signal superposition and computation accuracy [1], [3], [7].

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1      % This is a comment
2      x = linspace(0, 2*pi, 100);
3      y = sin(x) + 0.5 * randn(size(x)); % Numbers like 0.5 will be in Math Font
4      plot(x, y);

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1.c.ii Channel State Information (CSI) Imperfection

The need for accurate and fresh Channel State Information (CSI) at transmitters or receivers for many precoding/equalization techniques [1], [2], [7].

1.c.iii Waveform and Filter Design

Addressing time sampling error and intersymbol interference ISI inherent in digital communication transceivers when applied to OTA [7].

1.d Optimization Strategies to Enhance OAC Performance

- *Power Management*: Strategies to mitigate aggregation errors, such as optimizing power control policies to minimize Mean Squared Error MSE or maximize convergence speed [1], [4], [7].
- *Transceiver Design*: Joint design of pre-processing functions, precoders/beamformers, and post-processing functions (decoders) to match the channel to the desired function and minimize distortion [1], [2], [3], [8].

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